2026 03 19
발표 자료

광운대학교 로봇학과
FAIR Lab

김한서

KWANGWOON
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이번 주 진행사항

- Implicit Forecaster
    - 아이디어 적용 후 실험

Implicit Forecaster

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실험 세팅

- 사용한 모델: PatchTST + IF, iTransformer + IF
- 재현 실험 데이터셋: ETTh1, Weather

Experiment	ETTh1
Learning rate	10^{-4}
Epoch	100
Batch size	128
Loss function	MSE Loss
Seq_len	336
Pred_len	96/192/336/720
Layers	3
d_model	128
d_ff	512
Patience	10

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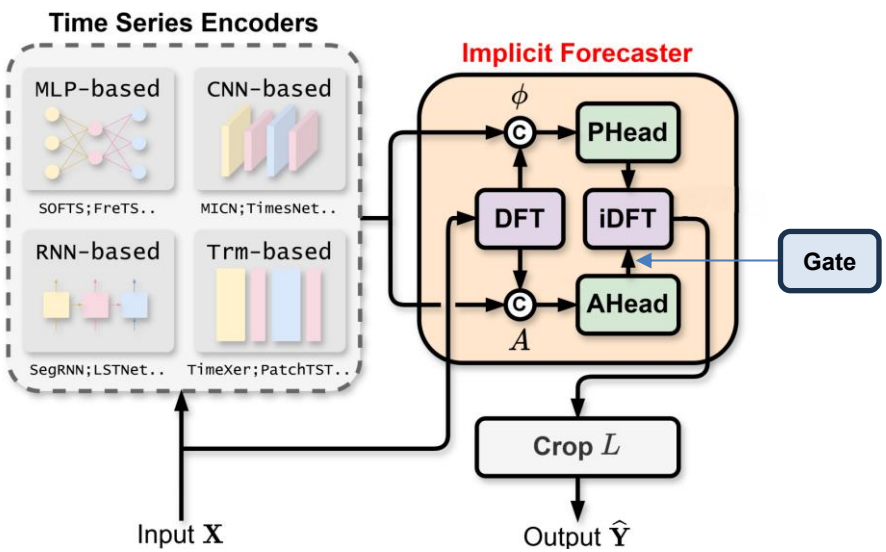
기존 아이디어



- Ahead의 예측 값을 Learnable Gate Head에 입력하여 진폭이 실제 신호인지 노이즈인지 학습 $g = \sigma\left(MLP_{gate}(\hat{A}_{raw})\right)$, $g \in [0,1]^{N \times P}$
- Sigmoid 함수를 통해 가중치 g로 변환, 기존 Ahead 예측 값과 g를 곱하여 최종 Amplitude 생성 $\hat{A}_{final} = \hat{A}_{raw} \odot g$
- $\mathcal{L}_{amp} = \mathrm{MSE}(\hat{A}_{final}, A_{GT})$, where $A_{GT} = |\mathcal{F}(Y)|$

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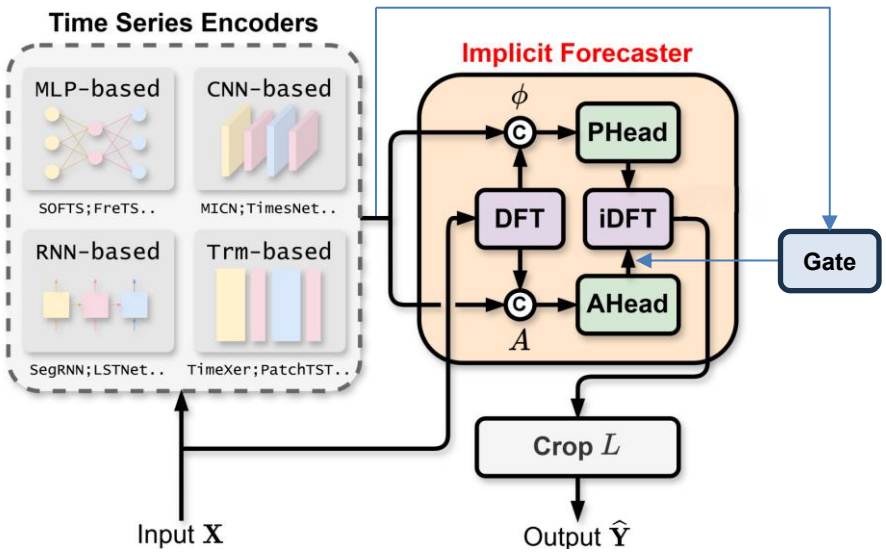
수정된 아이디어



- Gating Mechanism에 맞게 Ahead와 Gate Head의 입력을 동일하게 한 뒤에 실험했을 때, 성능 변화가 크지 않은 것을 확인
- Encoder의 출력 값을 Gate Head에 입력한 뒤 Sigmoid 함수를 통해 가중치 g로 변환, 이후 Ahead 예측 값과 가중치 g를 곱하여 최종 Amplitude 생성

$$g = \sigma\left(MLP_{gate}(X_{enc})\right), \quad g \in [0,1]^{N\times P}, \quad \hat{A}_{final} = \hat{A}_{raw} \odot g$$

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PatchTST + IF 재현 실험 (ETTh1)

- Gate Head 입력: Encoder 출력값, 원본 시계열 데이터의 Amplitude를 concat하여 사용

	ETTh1 Reproduction		ETTh1 Gate Head + MAE Loss 0.1		ETTh1 Gate Head + MAE Loss 0.0	
Pred len	MSE	MAE	MSE	MAE	MSE	MAE
96	**0.411**	0.454	0.412	0.452	**0.411**	**0.450**
192	0.454	**0.480**	**0.453**	**0.480**	0.456	0.481
336	**0.499**	**0.511**	0.515	0.520	0.515	0.520
720	0.633	0.591	**0.623**	**0.583**	**0.623**	**0.583**

Loss 0.1: 기존 MSE Loss에 $Loss_{amp}$를 10% 추가 반영한 것

Loss 0.0: 기존 MSE Loss만 사용한 것

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PatchTST + IF 재현 실험 (ETTh1)

- Gate Head 입력: Encoder 출력값만 사용

	ETTh1 Reproduction		ETTh1 Gate Head + MAE Loss 0.1		ETTh1 Gate Head + MAE Loss 0.0	
Pred len	MSE	MAE	MSE	MAE	MSE	MAE
96	0.411	0.454	**0.406**	**0.447**	0.409	0.449
192	0.454	0.480	**0.453**	**0.477**	0.456	0.481
336	0.499	0.511	0.502	**0.508**	**0.496**	**0.508**
720	0.633	0.591	0.627	0.587	**0.616**	**0.583**

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PatchTST + IF 재현 실험 (Weather)

- Gate Head 입력: Encoder 출력값만 사용

	Weather Reproduction		Weather Gate Head + MAE Loss 0.1		Weather Gate Head + MAE Loss 0.0	
Pred len	MSE	MAE	MSE	MAE	MSE	MAE
96	**0.158**	0.206	0.161	**0.200**	0.166	0.203
192	**0.196**	0.242	0.210	0.242	0.206	**0.240**
336	**0.249**	**0.278**	0.257	**0.278**	0.258	0.279
720	**0.312**	0.321	0.318	0.322	0.317	0.321

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iTransformer + IF 재현 실험 (ETTh1)

- Gate Head 입력: Encoder 출력값만 사용

	ETTh1 Reproduction		ETTh1 Gate Head + MAE Loss 0.1		ETTh1 Gate Head + MAE Loss 0.0	
Pred len	MSE	MAE	MSE	MAE	MSE	MAE
96	0.447	**0.462**	0.444	0.465	**0.442**	**0.462**
192	0.493	0.495	0.496	0.496	**0.491**	**0.492**
336	0.534	0.520	0.542	0.520	**0.533**	**0.518**
720	**0.662**	**0.595**	0.667	0.599	0.669	0.599

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iTransformer + IF 재현 실험 (Weather)

- Gate Head 입력: Encoder 출력값만 사용

	Weather Reproduction		Weather Gate Head + MAE Loss 0.1		Weather Gate Head + MAE Loss 0.0	
Pred len	MSE	MAE	MSE	MAE	MSE	MAE
96	0.164	0.202	0.153	0.195	0.159	0.199
192	0.207	0.240			0.205	0.241
336	0.259	0.280			0.251	0.277
720	0.319	0.322			0.313	0.319

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이후 계획

- Ahead 및 Phead
  - 다른 Loss로 변경 후 추가 실험

부록

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주요 모델 성능 비교

Models		SOFTS (2024)		w/ IF (Ours)		TimeXer (2024)		w/ IF (Ours)		iTransformer (2024)		w/ IF (Ours)		SegRNN (2023)		w/ IF (Ours)		PatchTST (2023)		w/ IF (Ours)		Crossformer (2023)		w/ IF (Ours)		TimesNet (2023)		w/ IF (Ours)	
Metric		MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE
ETTh1	96	0.442	0.461	**0.426**	**0.458**	0.423	**0.457**	**0.420**	0.461	**0.447**	**0.469**	0.456	0.474	**0.432**	**0.463**	0.436	0.467	0.425	0.457	**0.411**	**0.453**	**0.437**	**0.476**	0.437	**0.474**	0.484	0.501	**0.433**	**0.475**
	192	0.493	0.495	**0.469**	**0.489**	0.487	0.503	**0.461**	**0.494**	0.495	**0.503**	**0.490**	**0.503**	0.480	0.497	**0.467**	**0.490**	**0.477**	**0.494**	0.485	0.500	**0.487**	**0.505**	0.581	0.563	0.542	0.536	**0.498**	**0.521**
	336	0.529	**0.519**	**0.510**	0.523	0.529	**0.528**	**0.524**	0.541	0.537	**0.526**	**0.533**	0.532	0.524	0.526	**0.502**	**0.517**	0.552	0.530	**0.497**	**0.511**	**0.640**	0.602	0.644	**0.590**	0.606	0.576	**0.548**	**0.553**
	720	0.667	0.602	**0.618**	**0.595**	0.668	**0.612**	**0.637**	0.618	0.662	**0.601**	**0.660**	0.608	**0.659**	**0.608**	0.689	0.631	0.678	0.607	**0.601**	**0.582**	0.862	0.696	**0.684**	**0.630**	0.752	0.657	**0.662**	**0.637**
	Avg	0.533	0.519	**0.506**	**0.516**	0.527	**0.525**	**0.510**	0.528	**0.535**	**0.525**	**0.535**	0.529	0.524	**0.523**	**0.523**	0.526	0.533	0.522	**0.498**	**0.511**	0.607	0.570	**0.587**	**0.564**	0.596	0.568	**0.535**	**0.546**
ETTh2	96	0.159	0.279	**0.154**	**0.273**	0.153	0.273	**0.151**	**0.270**	0.160	0.280	**0.159**	**0.277**	0.149	**0.269**	**0.148**	0.270	0.152	0.272	**0.147**	**0.266**	0.179	0.310	**0.171**	**0.302**	0.168	0.290	**0.161**	**0.279**
	192	0.188	0.304	**0.182**	**0.295**	0.184	0.301	**0.177**	**0.292**	0.187	0.303	**0.186**	**0.300**	0.180	0.296	**0.175**	**0.295**	0.183	0.302	**0.179**	**0.295**	0.261	0.381	**0.223**	**0.353**	0.209	0.323	**0.199**	**0.309**
	336	0.208	0.320	**0.195**	**0.309**	0.215	0.327	**0.209**	**0.322**	0.208	0.322	**0.202**	**0.315**	0.200	0.315	**0.192**	**0.312**	0.198	0.316	**0.194**	**0.309**	0.256	0.384	**0.251**	**0.371**	0.216	0.332	**0.210**	**0.322**
	720	0.254	0.356	**0.240**	**0.342**	0.263	0.367	**0.239**	**0.341**	0.253	0.356	**0.244**	**0.346**	**0.242**	**0.350**	0.243	0.356	0.234	0.343	**0.227**	**0.333**	0.487	0.505	**0.306**	**0.413**	0.257	0.361	**0.246**	**0.349**
	Avg	0.202	0.315	**0.193**	**0.305**	0.204	0.317	**0.194**	**0.306**	0.202	0.315	**0.198**	**0.309**	0.193	**0.307**	**0.189**	0.308	0.192	0.308	**0.187**	**0.301**	0.296	0.395	**0.238**	**0.360**	0.212	0.327	**0.204**	**0.315**
ETTm1	96	**0.398**	**0.416**	0.406	0.425	**0.417**	**0.432**	0.435	0.438	**0.392**	0.426	0.396	**0.425**	0.400	0.424	**0.389**	**0.416**	**0.390**	0.418	0.403	**0.417**	0.497	0.488	**0.487**	**0.467**	0.487	0.469	**0.463**	**0.460**
	192	**0.438**	**0.448**	0.441	0.454	0.458	0.469	**0.446**	**0.458**	0.447	**0.458**	**0.443**	**0.458**	0.438	0.455	**0.435**	**0.450**	**0.445**	0.455	**0.445**	**0.452**	0.515	0.497	**0.479**	**0.480**	0.514	0.490	**0.478**	**0.476**
	336	0.520	0.492	**0.488**	**0.485**	0.509	0.503	**0.495**	**0.495**	0.523	0.498	**0.499**	**0.493**	**0.492**	0.489	0.495	**0.488**	0.504	0.490	**0.490**	**0.483**	0.541	0.524	**0.531**	**0.517**	0.573	0.534	**0.517**	**0.505**
	720	0.589	0.540	**0.555**	**0.537**	0.577	0.550	**0.534**	**0.529**	0.585	0.540	**0.572**	**0.538**	**0.548**	**0.529**	0.562	0.536	0.571	0.540	**0.544**	**0.527**	**0.600**	**0.554**	0.633	0.580	0.620	0.557	**0.572**	**0.547**
	Avg	0.486	**0.474**	**0.473**	0.475	0.490	0.488	**0.477**	**0.480**	0.487	0.481	**0.478**	**0.478**	**0.470**	0.474	**0.470**	**0.473**	0.477	0.476	**0.471**	**0.470**	0.538	0.516	**0.532**	**0.511**	0.548	0.512	**0.508**	**0.497**
ETTm2	96	0.109	0.227	**0.101**	**0.220**	**0.098**	**0.217**	0.099	**0.217**	0.107	0.227	**0.105**	**0.225**	**0.097**	**0.216**	0.100	0.217	0.102	0.222	**0.100**	**0.219**	0.112	0.240	**0.105**	**0.227**	0.106	0.225	**0.101**	**0.218**
	192	0.130	0.252	**0.128**	**0.249**	0.125	0.247	**0.124**	**0.244**	0.132	0.255	**0.130**	**0.253**	**0.123**	**0.244**	0.125	0.246	0.128	0.251	**0.127**	**0.250**	0.147	0.267	**0.137**	**0.262**	0.136	0.254	**0.130**	**0.248**
	336	0.162	0.280	**0.158**	**0.279**	**0.155**	**0.273**	0.156	**0.273**	0.168	0.286	**0.161**	**0.280**	**0.151**	**0.270**	0.154	0.274	0.158	0.279	**0.152**	**0.272**	0.238	0.344	**0.209**	**0.338**	0.165	0.281	**0.158**	**0.276**
	720	0.204	0.315	**0.202**	**0.314**	0.201	0.311	**0.200**	**0.309**	0.209	0.319	**0.205**	**0.315**	**0.195**	**0.306**	0.198	0.313	0.199	0.310	**0.195**	**0.308**	0.315	0.425	**0.282**	**0.391**	0.215	0.322	**0.203**	**0.313**
	Avg	0.151	0.269	**0.147**	**0.266**	**0.145**	0.262	**0.145**	**0.261**	0.154	0.272	**0.150**	**0.268**	**0.142**	**0.259**	0.144	0.263	0.147	0.266	**0.144**	**0.262**	0.203	0.319	**0.183**	**0.304**	0.155	0.271	**0.148**	**0.264**
ECL	96	0.146	**0.233**	**0.141**	**0.233**	0.142	0.242	**0.137**	**0.231**	0.150	**0.239**	**0.145**	**0.239**	**0.157**	**0.249**	0.165	0.254	0.179	0.271	**0.163**	**0.250**	0.164	0.266	**0.161**	**0.263**	0.167	0.268	**0.160**	**0.265**
	192	**0.157**	**0.245**	**0.157**	0.250	0.155	0.252	**0.151**	**0.247**	0.160	**0.248**	**0.158**	0.253	**0.169**	**0.260**	0.174	0.263	0.184	0.275	**0.170**	**0.259**	0.186	0.284	**0.176**	**0.277**	0.182	0.282	**0.178**	**0.281**
	336	0.170	**0.259**	**0.166**	0.263	0.171	0.271	**0.165**	**0.261**	0.172	**0.263**	**0.169**	0.268	**0.181**	**0.277**	0.185	0.278	0.197	0.289	**0.182**	**0.274**	0.213	0.308	**0.200**	**0.303**	**0.192**	**0.293**	0.194	0.298
	720	0.208	**0.294**	**0.197**	0.296	0.198	0.295	**0.190**	**0.290**	0.208	**0.295**	**0.196**	**0.295**	**0.216**	**0.312**	0.219	**0.312**	0.234	0.321	**0.221**	**0.314**	0.285	0.376	**0.236**	**0.335**	**0.218**	**0.315**	**0.218**	0.320
	Avg	0.170	**0.258**	**0.165**	0.261	0.166	0.265	**0.161**	**0.257**	0.172	**0.261**	**0.167**	0.264	**0.181**	**0.275**	0.186	0.277	0.199	0.289	**0.184**	**0.274**	0.212	0.308	**0.193**	**0.294**	0.190	**0.289**	**0.188**	0.291
Traffic	96	0.393	**0.267**	**0.390**	**0.267**	**0.438**	**0.278**	0.690	0.341	**0.401**	**0.277**	0.432	0.302	0.609	0.313	**0.530**	**0.298**	0.463	0.304	**0.439**	**0.293**	0.544	**0.311**	**0.512**	0.319	0.579	**0.322**	**0.573**	0.348
	192	0.426	0.281	**0.420**	**0.280**	**0.471**	**0.296**	0.664	0.347	**0.427**	**0.288**	0.458	0.311	0.626	0.319	**0.559**	**0.306**	0.478	0.311	**0.460**	**0.302**	0.565	**0.320**	**0.525**	0.329	0.608	**0.332**	**0.584**	0.363
	336	0.444	**0.289**	**0.440**	0.291	**0.487**	**0.298**	0.621	0.347	**0.441**	**0.294**	0.482	0.324	0.644	0.326	**0.571**	**0.316**	0.493	0.317	**0.475**	**0.308**	0.579	**0.328**	**0.548**	0.337	0.622	**0.338**	**0.592**	0.350
	720	0.468	**0.307**	**0.467**	0.309	**0.523**	**0.316**	0.541	0.338	**0.467**	**0.310**	0.617	0.396	0.675	0.345	**0.633**	**0.339**	0.520	0.335	**0.507**	**0.331**	0.610	**0.346**	**0.598**	0.358	0.648	**0.351**	**0.623**	0.376
	Avg	0.433	**0.286**	**0.429**	0.287	**0.480**	**0.297**	0.629	0.343	**0.434**	**0.292**	0.497	0.333	0.639	0.326	**0.573**	**0.315**	0.489	0.317	**0.470**	**0.308**	0.575	**0.326**	**0.546**	0.336	0.614	**0.336**	**0.593**	0.359
Weather	96	0.164	**0.205**	**0.161**	0.214	0.147	**0.194**	**0.145**	0.196	0.164	0.202	**0.154**	**0.201**	**0.156**	**0.217**	0.160	0.218	0.164	**0.205**	**0.158**	0.206	**0.163**	**0.230**	0.164	0.233	0.165	0.213	**0.156**	**0.210**
	192	**0.204**	**0.240**	0.207	0.257	0.191	**0.234**	**0.189**	0.238	0.207	**0.240**	**0.198**	0.242	**0.198**	0.258	0.199	**0.255**	0.207	**0.242**	**0.196**	**0.242**	**0.222**	0.295	0.224	**0.292**	0.207	**0.249**	**0.203**	0.250
	336	0.257	0.278	**0.241**	**0.277**	**0.244**	**0.273**	0.247	0.282	0.259	0.280	**0.244**	**0.277**	0.251	0.300	**0.245**	**0.294**	0.258	0.279	**0.249**	**0.278**	**0.256**	**0.320**	0.263	0.323	0.272	0.293	**0.255**	**0.289**
	720	0.319	**0.322**	**0.309**	0.331	0.304	**0.314**	**0.303**	0.324	0.319	**0.322**	**0.305**	0.323	0.317	0.348	**0.305**	**0.340**	0.320	**0.320**	**0.312**	0.321	**0.327**	**0.377**	0.359	0.404	0.322	0.327	**0.309**	**0.326**
	Avg	0.236	**0.261**	**0.229**	0.270	**0.221**	**0.254**	**0.221**	0.260	0.237	**0.261**	**0.225**	**0.261**	0.230	0.281	**0.227**	**0.277**	0.237	**0.262**	**0.229**	**0.262**	**0.242**	**0.305**	0.253	0.313	0.241	0.270	**0.231**	**0.269**
Exchange	96	0.089	0.214	**0.082**	**0.204**	0.086	0.207	**0.082**	**0.204**	0.103	0.231	**0.102**	**0.229**	**0.082**	**0.200**	0.085	0.208	0.091	0.212	**0.086**	**0.206**	**0.236**	**0.352**	0.262	0.372	0.096	0.225	**0.090**	**0.217**
	192	0.200	0.328	**0.177**	**0.305**	0.187	0.312	**0.177**	**0.306**	0.206	0.334	**0.188**	**0.318**	**0.174**	**0.298**	0.207	0.333	0.184	0.310	**0.183**	**0.309**	**0.427**	**0.484**	0.526	0.539	**0.185**	**0.319**	0.187	0.321
	336	0.389	0.465	**0.349**	**0.437**	**0.371**	**0.449**	0.373	0.452	**0.362**	**0.448**	0.364	0.450	0.360	0.437	**0.306**	**0.424**	0.361	0.445	**0.345**	**0.434**	**1.063**	**0.807**	2.276	1.191	**0.381**	**0.456**	0.382	0.460
	720	1.017	0.809	**0.999**	**0.807**	**0.943**	**0.763**	0.950	0.773	1.038	0.820	**0.994**	**0.798**	1.142	0.848	**0.578**	**0.608**	1.128	0.847	**1.074**	**0.834**	**1.466**	**0.973**	2.287	1.198	1.037	0.825	**1.023**	**0.816**
	Avg	0.424	0.454	**0.402**	**0.438**	0.397	**0.433**	**0.395**	0.434	0.427	0.458	**0.412**	**0.449**	0.440	0.446	**0.294**	**0.393**	0.441	0.454	**0.422**	**0.446**	**0.798**	**0.654**	1.338	0.825	0.425	0.456	**0.420**	**0.454**
ILI	96	2.731	**1.001**	**2.621**	1.035	**2.903**	1.134	2.941	**1.124**	2.820	1.085	**2.543**	**1.014**	3.974	1.272	**2.920**	**1.078**	2.585	1.005	**2.292**	**1.003**	**3.922**	**1.375**	4.598	1.508	**2.715**	**1.054**	2.811	1.057
	192	**2.665**	**1.105**	2.837	1.121	3.052	1.153	**2.677**	**1.119**	3.034	1.173	**2.880**	**1.113**	4.366	1.379	**3.167**	**1.173**	4.506	1.292	**4.131**	**1.252**	**3.971**	**1.383**	5.034	1.573	**3.670**	**1.245**	5.866	1.333
	336	3.485	1.280	**3.212**	**1.212**	3.402	1.235	**2.926**	**1.154**	**3.158**	**1.224**	3.263	1.232	4.425	1.427	**3.932**	**1.339**	**3.578**	1.219	3.736	**1.202**	**3.958**	**1.420**	4.861	1.543	3.633	1.229	**3.537**	**1.228**
	720	2.691	**1.101**	**2.637**	1.111	**3.179**	**1.174**	3.208	1.191	2.953	1.175	**2.797**	**1.116**	5.759	1.651	**4.547**	**1.395**	2.792	1.102	**2.581**	**1.050**	**5.132**	**1.582**	6.021	1.745	3.037	1.120	**2.723**	**1.089**
	Avg	2.893	1.122	**2.827**	**1.120**	3.134	1.174	**2.938**	**1.147**	2.991	1.164	**2.871**	**1.119**	4.631	1.432	**3.641**	**1.246**	3.365	1.155	**3.185**	**1.127**	**4.246**	**1.440**	5.128	1.592	**3.264**	**1.162**	3.734	1.177
PEMS03	12	0.064	0.167	**0.061**	**0.164**	0.064	0.177	**0.058**	**0.165**	0.065	0.172	**0.062**	**0.169**	0.078	0.191	**0.074**	**0.185**	0.092	0.208	**0.070**	**0.178**	0.083	0.194	**0.073**	**0.185**	0.087	0.195	**0.069**	**0.173**
	24	0.081	**0.189**	**0.080**	**0.189**	0.086	0.203	**0.072**	**0.181**	0.093	0.206	**0.085**	**0.197**	0.123	0.243	**0.113**	**0.229**	0.144	0.266	**0.105**	**0.218**	0.117	0.237	**0.104**	**0.224**	0.118	0.222	**0.094**	**0.192**
	48	**0.118**	**0.230**	0.127	0.238	0.130	0.248	**0.110**	**0.220**	**0.160**	**0.273**	0.279	0.360	0.225	0.333	**0.189**	**0.301**	0.275	0.372	**0.177**	**0.285**	0.207	0.321	**0.174**	**0.287**	0.186	0.272	**0.119**	**0.220**
	96	**0.165**	**0.272**	0.167	0.275	0.207	0.311	**0.167**	**0.269**	1.337	0.918	**0.410**	**0.447**	0.356	0.425	**0.261**	**0.364**	0.496	0.518	**0.285**	**0.370**	0.260	0.363	**0.234**	**0.340**	0.309	0.352	**0.142**	**0.248**
	Avg	**0.107**	**0.214**	0.109	0.216	0.122	0.235	**0.102**	**0.209**	0.414	0.392	**0.209**	**0.293**	0.196	0.298	**0.159**	**0.270**	0.252	0.341	**0.159**	**0.263**	0.167	0.279	**0.146**	**0.259**	0.175	0.260	**0.106**	**0.208**
PEMS04	12	**0.080**	**0.185**	0.085	0.193	0.079	0.192	**0.072**	**0.175**	0.094	0.199	**0.084**	**0.189**	0.103	0.215	**0.097**	**0.208**	0.123	0.242	**0.096**	**0.205**	0.095	0.210	**0.088**	**0.200**	0.135	0.251	**0.082**	**0.190**
	24	0.102	0.213	**0.093**	**0.199**	0.095	0.212	**0.085**	**0.193**	0.130	0.236	**0.102**	**0.211**	0.157	0.273	**0.144**	**0.258**	0.196	0.309	**0.139**	**0.252**	0.126	0.248	**0.111**	**0.230**	0.180	0.290	**0.095**	**0.208**
	48	0.125	0.238	**0.116**	**0.227**	0.120	0.241	**0.100**	**0.211**	0.197	0.296	**0.134**	**0.247**	0.265	0.363	**0.225**	**0.330**	0.371	0.430	**0.217**	**0.321**	0.176	0.300	**0.143**	**0.264**	0.240	0.342	**0.125**	**0.243**
	96	0.141	0.254	**0.136**	**0.247**	0.157	0.279	**0.136**	**0.250**	0.328	0.398	**0.178**	**0.288**	0.406	0.462	**0.298**	**0.390**	0.647	0.591	**0.294**	**0.384**	0.529	0.567	**0.184**	**0.301**	0.335	0.418	**0.166**	**0.288**
	Avg	0.112	0.223	**0.107**	**0.216**	0.113	0.231	**0.098**	**0.207**	0.187	0.282	**0.124**	**0.234**	0.233	0.328	**0.191**	**0.296**	0.334	0.393	**0.186**	**0.290**	0.232	0.331	**0.132**	**0.249**	0.222	0.325	**0.117**	**0.232**
PEMS07	12	0.067	**0.163**	**0.062**	**0.163**	0.064	0.169	**0.055**	**0.150**	0.067	0.167	**0.060**	**0.158**	0.079	0.187	**0.073**	**0.176**	0.089	0.209	**0.072**	**0.176**	0.097	0.204	**0.076**	**0.188**	0.105	0.222	**0.068**	**0.169**
	24	0.083	0.185	**0.074**	**0.174**	0.084	0.193	**0.069**	**0.170**	0.098	0.201	**0.081**	**0.185**	0.130	0.244	**0.117**	**0.227**	0.152	0.270	**0.116**	**0.225**	0.141	0.252	**0.113**	**0.227**	0.130	0.248	**0.090**	**0.191**
	48	0.103	**0.204**	**0.102**	0.207	0.108	0.212	**0.080**	**0.181**	1.511	1.012	**0.205**	**0.309**	0.241	0.338	**0.206**	**0.306**	0.303	0.385	**0.201**	**0.297**	0.255	0.331	**0.178**	**0.282**	0.179	0.294	**0.116**	**0.227**
	96	**0.130**	**0.230**	0.131	0.237	0.142	0.247	**0.107**	**0.210**	1.860	1.138	**0.250**	**0.346**	0.393	0.439	**0.300**	**0.380**	0.556	0.539	**0.302**	**0.370**	0.291	0.374	**0.255**	**0.339**	0.253	0.355	**0.161**	**0.273**
	Avg	0.096	**0.195**	**0.092**	**0.195**	0.100	0.205	**0.078**	**0.178**	0.884	0.629	**0.149**	**0.249**	0.211	0.302	**0.174**	**0.272**	0.275	0.351	**0.173**	**0.267**	0.196	0.290	**0.155**	**0.259**	0.167	0.280	**0.109**	**0.215**
PEMS08	12	0.076	0.179	**0.070**	**0.173**	0.093	0.211	**0.074**	**0.184**	0.087	0.193	**0.078**	**0.184**	0.093	0.203	**0.085**	**0.192**	0.102	0.218	**0.082**	**0.188**	0.098	0.205	**0.085**	**0.195**	0.125	0.240	**0.081**	**0.189**
	24	0.107	0.211	**0.098**	**0.207**	0.134	0.256	**0.101**	**0.217**	0.132	0.242	**0.110**	**0.220**	0.145	0.258	**0.127**	**0.240**	0.168	0.279	**0.122**	**0.231**	0.145	0.260	**0.116**	**0.231**	0.161	0.277	**0.098**	**0.210**
	48	0.153	0.248	**0.123**	**0.230**	0.140	0.256	**0.100**	**0.208**	1.649	1.082	**0.228**	**0.335**	0.253	0.351	**0.202**	**0.311**	0.321	0.406	**0.199**	**0.305**	0.197	0.311	**0.153**	**0.271**	0.219	0.326	**0.134**	**0.251**
	96	**0.136**	**0.236**	0.137	0.237	0.210	0.318	**0.156**	**0.261**	1.571	0.986	**0.302**	**0.397**	0.437	0.467	**0.290**	**0.385**	0.614	0.581	**0.287**	**0.379**	0.439	0.493	**0.216**	**0.331**	0.335	0.416	**0.199**	**0.318**
	Avg	0.118	0.218	**0.107**	**0.212**	0.144	0.260	**0.108**	**0.217**	0.860	0.626	**0.179**	**0.284**	0.232	0.320	**0.176**	**0.282**	0.301	0.371	**0.172**	**0.276**	0.220	0.317	**0.142**	**0.257**	0.210	0.315	**0.128**	**0.242**
Solar	96	0.192	0.233	**0.185**	**0.231**	0.207	0.268	**0.202**	**0.250**	0.210	0.253	**0.202**	**0.243**	0.226	0.290	**0.218**	**0.273**	0.211	0.271	**0.210**	**0.256**	**0.193**	**0.257**	**0.193**	0.259	0.211	**0.236**	**0.198**	0.253
	192	0.223	**0.253**	**0.215**	0.255	0.241	0.293	**0.240**	**0.276**	0.240	0.277	**0.225**	**0.268**	0.244	0.304	**0.240**	**0.288**	0.247	0.294	**0.235**	**0.272**	0.231	**0.281**	**0.224**	0.287	0.256	0.281	**0.221**	**0.274**
	336	0.238	0.263	**0.218**	**0.262**	0.257	0.303	**0.243**	**0.282**	0.258	0.291	**0.246**	**0.287**	0.259	0.304	**0.248**	**0.291**	0.257	0.296	**0.237**	**0.281**	0.249	0.298	**0.236**	**0.293**	0.288	0.282	**0.222**	**0.280**
	720	0.234	0.264	**0.210**	**0.263**	0.251	0.297	**0.237**	**0.281**	0.255	0.291	**0.238**	**0.285**	0.249	0.295	**0.245**	**0.291**	0.246	0.293	**0.233**	**0.278**	0.522	0.539	**0.235**	**0.294**	0.285	0.295	**0.218**	**0.271**
	Avg	0.222	**0.253**	**0.207**	**0.253**	0.239	0.290	**0.231**	**0.272**	0.241	0.278	**0.228**	**0.271**	0.244	0.298	**0.238**	**0.286**	0.240	0.288	**0.229**	**0.272**	0.299	0.344	**0.222**	**0.283**	0.260	0.273	**0.212**	**0.270**

Implicit Forecaster

FAIR Lab

KWANGWOON UNIVERSITY

PatchTST + IF 재현 실험 (ETTh1)

	ETTh1 Paper		ETTh1 Reproduction		ETTh1 Gate Head + MAE Loss 0.1		ETTh1 Gate Head + MAE Loss 0.0	
Pred len	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE
96	0.411	0.453	0.411	0.454	0.406	0.447	0.409	0.449
192	0.485	0.500	0.454	0.480	0.453	0.477	0.456	0.481
336	0.497	0.511	0.499	0.511	0.502	0.508	0.496	0.508
720	0.601	0.582	0.633	0.591	0.627	0.587	0.616	0.583

Implicit Forecaster

FAIR Lab

KWANGWOON UNIVERSITY

iTransformer + IF 재현 실험 (ETTh1)

	ETTh1 Paper		ETTh1 Reproduction		ETTh1 Gate Head + MAE Loss 0.1		ETTh1 Gate Head + MAE Loss 0.0	
Pred len	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE
96	0.456	0.474	0.447	0.462	0.444	0.465	0.442	0.462
192	0.490	0.503	0.493	0.495	0.496	0.496	0.491	0.492
336	0.533	0.532	0.534	0.520	0.542	0.520	0.533	0.518
720	0.660	0.608	0.662	0.595	0.667	0.599	0.669	0.599